Radar HRRP Statistical Recognition: Parametric Model and Model Selection

Lan Du, Hongwei Liu, *Member, IEEE*, and Zheng Bao, *Senior Member, IEEE*

***Abstract*—Statistical modeling for radar high-resolution range profile (HRRP) is a challenging task in radar HRRP statistical recognition. Theoretical analysis and experimental results show that elements in an HRRP sample are statistically correlated and non-Gaussian distributed. First, this paper introduces three joint-Gaussian models, i.e., subspace approximation model, probability principal components analysis (PPCA) model and factor analysis (FA) model, into radar HRRP statistical recognition. Due to the experimental results, we can have the conclusion that the jointly non-Gaussian distributed HRRP samples approximately follow the joint-Gaussian distribution described by FA model. Therefore, we can apply FA model to radar HRRP statistical recognition rather than a joint-Gaussian mixture model, e.g., PPCA mixture model or FA mixture model, which is a more accurate choice for modeling non-Gaussian distributed correlations in multidimensional data but with high learning complexity and large computation burden, and the difficulty in the statistical modeling for HRRP samples is largely reduced. Second, this paper concerns model selection of FA model in radar HRRP statistical recognition, in which there are two issues, i.e., the partition of target-aspect frames and the determination of the number of factors in each frame. Based on the Akaike information criterion (AIC) and the Bayes' information criterion (BIC), an iterated algorithm for model selection is proposed in this paper, which can automatically give the optimal aspect-frame boundaries and determine the optimal number of factors in each aspect-frame. The recognition experiments based on measured data show that the proposed adaptive partition approach can further improve the recognition performance with higher recognition efficiency.**

***Index Terms*—Akaike information criterion (AIC), Bayesian information criterion (BIC), factor analysis (FA), high-resolution range profile (HRRP), model selection, probability principal components analysis (PPCA), radar automatic target recognition (RATR), statistical recognition with parametric model.**

I. INTRODUCTION

A HIGH-RESOLUTION range profile (HRRP) is the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS). It contains the target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community [1]–[17]. Some literature [13]–[15] showed that statistical recognition is an efficient method for RATR. By statistical recognition is meant that the class membership of a measurement HRRP sample $\boldsymbol{x}$ is determined based on the posterior probabilities of all class memberships $P(k \mid \boldsymbol{x})$ ($k = 1, 2, \ldots, K$ denotes class membership). According to Bayes' theorem, we have $P(k \mid \boldsymbol{x}) \propto p(\boldsymbol{x} \mid k)P(k)$, where $p(\boldsymbol{x} \mid k)$ is the class-conditional probability density and $P(k)$ is the prior class probability. Thus, estimation of the posterior probability is turned into estimation of the class-conditional density. For an HRRP sample including many range cells, nonparametric methods of density estimation, e.g., Parzen window method, are impractical due to the huge storage requirement. An alternative approach is to trade flexibility for robustness and to use some simple parametric model for target probability density. Under the hypothesis that elements in an HRRP sample are statistically independent, some simple statistical models, i.e., Gaussian model [6], Gamma model [13], [14], and the two-distribution compounded model comprising Gamma distribution and Gaussian mixture distribution [15], were proposed for HRRP samples. However, theoretical analysis and experimental results based on measured data show that the independence assumption is not true. Motivated by this, this paper will focus on building a more accurate statistical model for HRRP samples under the prerequisite that the echoes from different range cells in an HRRP sample are statistically correlated.

If we assume that the d-dimensional HRRP samples in an aspect-frame follow a joint-Gaussian distribution, i.e., $\boldsymbol{x} \in \Re^d \sim \mathcal{N}(\boldsymbol{m_x}, \boldsymbol{C_x})$, three parametric models are introduced into radar HRRP statistical recognition: a) subspace approximation model; b) probability principal components analysis (PPCA) model [19]; and c) factor analysis (FA) model [20], [21], [26]. According to the scattering center model [18], which is a first-order approximation model without considering multiple reflections between scatterers and has been proved to be a simple and efficient target model in inverse synthetic aperture radar (ISAR) imaging and synthetic aperture radar (SAR) imaging, the echo from each range cell in an HRRP sample departs from Gaussian distribution, thus HRRP samples cannot strictly follow a joint-Gaussian distribution. However, from the recognition experiments and the experiments with the Kullback–Leibler (KL) distance between the probability density function (pdf) estimated by the Parzen window method for measured HRRP samples, referred to as the measured pdf in this paper, and the pdf assumed by a parametric model, referred to as the assumed pdf in this paper, used as the exactness criterion

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L. Du and Z. Bao are with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China.

H. Liu was with the Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708 USA. He is now with the National Laboratory of Radar Signal Processing, Xidian University, Xi'an, China.

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for a parametric model, we know that FA model can exactly describe HRRP samples' statistical characteristics and achieve good recognition performance in statistical recognition, which demonstrates that a jointly non-Gaussian distributed HRRP frame approximately follows the joint-Gaussian distribution described by FA model. Therefore, we can apply FA model to radar HRRP statistical recognition rather than a joint-Gaussian mixture model, e.g., PPCA mixture model or FA mixture model, which is a more accurate choice for modeling non-Gaussian distributed correlations in multidimensional data but with high learning complexity and large computation burden, and the difficulty in statistical modeling for HRRP samples is greatly reduced.

Then this paper concerns model selection of FA model in radar HRRP statistical recognition. Usually, model selection of FA model only deals with how to automatically determine the optimal number of factors. According to the scattering center model, an HRRP sample is very sensitive to the target-aspect variation. To deal with the target-aspect sensitivity, the literature [2], [3], [9], [12], [15], [16] determined the number of aspect-frames in the whole training data according to a target-aspect sector without the scatterers' motion through range cells (MTRC) and divided the training data into target-aspect frames at equal interval, which is referred to as the equal interval partition approach in this paper. Nevertheless, from the viewpoint of statistical recognition, the equal interval partition approach is not optimal. Therefore, how to divide aspect-frames for the training data is another important issue in model selection for radar HRRP statistical recognition. Based on the two most popular model selection criteria, i.e., the Akaike information criterion (AIC) and the Bayes information criterion (BIC), an iterated algorithm for model selection is proposed in this paper, which can automatically give the optimal aspect-frame boundaries and determine the optimal number of factors in each aspect-frame. The recognition experiments based on measured data show that the proposed adaptive partition approach can further improve the recognition performance.

This paper is organized as follows. In Section II, how to accurately describe HRRP samples' statistical characteristics is discussed. The subspace approximation model, PPCA model, and FA model are introduced into radar HRRP statistical recognition and compared by some experiments. In Section III, an iterated algorithm based on AIC and BIC for model selection of FA model in radar HRRP statistical recognition is proposed. Finally, conclusions are made in Section IV.

II. Statistical Modeling in Radar HRRP Statistical Recognition

As discussed in the literature [9], [15], [16], it is a prerequisite for radar HRRP target recognition to deal with the target-aspect, time-shift and amplitude-scale sensitivity problems. To deal with the target-aspect sensitivity, as discussed in Section I, the equal interval partition approach can be adopted despite of not an optimal one [15]. To deal with the time-shift sensitivity, time-shift compensation techniques in ISAR imaging, such as the slide correlation processing or its modifications [18], can be used in both of the training phase and test phase [15], [16]. To deal with the amplitude-scale sensitivity, the L_2 amplitude normalization can be used as a preprocessing procedure [15]. Thus, we will discuss that how to accurately describe a time-shift compensated and amplitude L_2 normalized HRRP frame's statistical characteristics in the following.

A. Statistical Correlation Property of HRRP Samples

The scattering center model of a target will vary with the target-aspect, because the variation of target-aspect may cause some scatterers to remove from one range cell to another, which is referred to as scatterers' MTRC. The limitation of target-aspect change for avoiding scatterers' MTRC is given by [18]

$$\delta\varphi \leq (\delta\varphi)_{\mathrm{MTRC}} \triangleq \frac{C}{2BL_x} \tag{1}$$

where B, L_x, and C are the bandwidth of the radar signal after range compression, the maximum target dimension in cross range and the speed of light, respectively. As the aforementioned discussion, the equal interval partition approach divides the aspect-frames just according to (1). Because the scatterers are localized immovably in each range cell within an aspect-frame, of which the distribution is irrelevant to that in another cell, some literature [6], [13]–[15] assumed that the echoes from different range cells within an aspect-frame are independent. Actually, this independence assumption is not true. There are several reasons.

1) According to the scattering center model, the echo from a range cell varies with the distance differences between the scatterers in this cell. When a target rotates, the scatterers from different range cells will rotate in the same way, which causes that the scatterers' distance differences in different cells dependently change, therefore, the statistical properties of the echoes from different cells are also dependent to some extent.
2) The returned echoes are multiplied by window functions before getting HRRP samples (e.g., the Hamming window was applied for the measured data in this paper) in order to compress the range sidelobe, therefore, the range resolution decreases and the echoes from two neighboring range cells (not those from the real range cells defined in Section I) in an HRRP sample are not completely independent.
3) Different from what assumed in the scattering center model, some ISAR images using measured data show that the multiple reflections phenomena exist. When the multiple reflections phenomenon occurs between the scatterers in two range cells, the returned echoes from the two cells cannot be independent.
4) The similarity in geometrical structure and scattering property between different parts of a target can also cause the echoes from different range cells to be statistically correlated.

Therefore, this paper will focus on building a more accurate statistical model for HRRP samples under the prerequisite that the echoes from different range cells in an HRRP sample are statistically correlated.

B. Parametric Models

We introduce three joint-Gaussian models, i.e., subspace approximation model, PPCA model and FA model, into radar HRRP statistical recognition, then make an analysis on the statistical descriptions' exactnesses and recognition performances of these parametric models.

1) Subspace Approximation Model: Assume that a d-dimensional HRRP sample in the lth $(l = 1, 2, \ldots, L_k)$ aspect-frame of the target $T_k (k = 1, 2, \ldots, K)$ follows a joint-Gaussian distribution, i.e., $\boldsymbol{x} \in \Re^d \sim \mathcal{N}(\boldsymbol{m}_{\boldsymbol{x}}, \boldsymbol{C}_{\boldsymbol{x}})$,

$$p(\boldsymbol{x} \mid l; k) = (2\pi)^{-dd/2} |\boldsymbol{C}_{kl}|^{-1/2} \times \exp\left\{-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{m}_{kl})^T \boldsymbol{C}_{kl}^{-1} (\boldsymbol{x} - \boldsymbol{m}_{kl})\right\}. \quad (2)$$

Let $\boldsymbol{X}_{kl} = \left\{\boldsymbol{x}_{kli} \in \Re^d \mid i = 1, 2, \ldots, N\right\}$ denote the set of training data. The log-likelihood of the training dataset under the distribution model given by (2) is

$$\begin{aligned} L(\boldsymbol{\Theta}_{kl}) &= \sum_{i=1}^{N} \ln\{p(\boldsymbol{x}_{kli})\} \\ &= -\frac{N}{2}\left\{d\ln(2\pi) + \ln(|\boldsymbol{C}_{kl}|) + \operatorname{tr}\left\{\boldsymbol{C}_{kl}^{-1}\left[\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{x}_{kli} - \boldsymbol{m}_{kl})(\boldsymbol{x}_{kli} - \boldsymbol{m}_{kl})^T\right]\right\}\right\} \end{aligned} \quad (3)$$

where $\boldsymbol{\Theta}_{kl} = \{\boldsymbol{m}_{kl}, \boldsymbol{C}_{kl}\}$ denotes the unknown parameters in this model. According to (3), the maximum-likelihood (ML) estimators of $\boldsymbol{m}_{kl}$ and $\boldsymbol{C}_{kl}$ are, respectively

$$\begin{aligned} \tilde{\boldsymbol{m}}_{kl} &= \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{x}_{kli}, \\ \tilde{\boldsymbol{C}}_{kl} &= \boldsymbol{S}_{kl} = \frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{x}_{kli} - \tilde{\boldsymbol{m}}_{kl})(\boldsymbol{x} - \tilde{\boldsymbol{m}}_{kl})^T \\ &= \boldsymbol{U}_{kl(d)}\boldsymbol{\Lambda}_{kl(d)}\boldsymbol{U}_{kl(d)}^T \approx \boldsymbol{U}_{kl(q)}\boldsymbol{\Lambda}_{kl(q)}\boldsymbol{U}_{kl(q)}^T \end{aligned} \quad (4)$$

where $\boldsymbol{S}_{kl}$ denotes the sample covariance matrix, $\boldsymbol{U}_{kl(d)} = [\boldsymbol{u}_{kl1} \ \ \boldsymbol{u}_{kl2} \ \ \ldots \ \ \boldsymbol{u}_{kld}]$ and $\boldsymbol{\Lambda}_{kl(d)} = \operatorname{diag}([\lambda_{kl1} \ \ \lambda_{kl2} \ \ \ldots \ \ \lambda_{kld}]^T)$, respectively, denote the corresponding eigenvector matrix and eigenvalue matrix of $\boldsymbol{S}_{kl}$ with $\lambda_{kl1} \geq \lambda_{kl2} \geq \cdots \geq \lambda_{kld}$ and $\operatorname{diag}(\boldsymbol{w})$ denoting the diagonal matrix in which the vector $\boldsymbol{w}$ is the diagonal. If we decompose the full covariance matrix into the signal subspace and the noise subspace, then $\boldsymbol{U}_{kl(d)} = [\boldsymbol{U}_{kl(q)} \quad \boldsymbol{U}_{kl(d-q)}]$ and

$$\boldsymbol{\Lambda}_{kl(d)} = \begin{bmatrix} \boldsymbol{\Lambda}_{kl(q)} & \\ & \boldsymbol{\Lambda}_{kl(d-q)} \end{bmatrix}$$

with $q < d$. In order to denoise, the signal subspace substitutes for the whole space in (4). Substituting the ML estimators given by (4) into (2)

$$\begin{aligned} p(\boldsymbol{x} \mid l; k) &= (2\pi)^{-d/2}\left|\tilde{\boldsymbol{C}}_{kl}\right|^{-1/2} \times \exp\left\{-\frac{1}{2}(\boldsymbol{x} - \tilde{\boldsymbol{m}}_{kl})^T \tilde{\boldsymbol{C}}_{kl}^{-1}(\boldsymbol{x} - \tilde{\boldsymbol{m}}_{kl})\right\} \\ &= (2\pi)^{-d/2}\left|\boldsymbol{\Lambda}_{kl(q)}\right|^{-1/2} \times \exp\left\{-\frac{1}{2}(\boldsymbol{x} - \tilde{\boldsymbol{m}}_{kl})^T \times \boldsymbol{U}_{kl(q)}\boldsymbol{\Lambda}_{kl(q)}^{-1}\boldsymbol{U}_{kl(q)}^T(\boldsymbol{x} - \tilde{\boldsymbol{m}}_{kl})\right\}. \end{aligned} \quad (5)$$

Based on Bayes' theorem, a test HRRP sample should belong to the class which has the maximum posterior probability given the test sample [6], [13]–[15]. Under the limit condition that $q = d$, we can get the full covariance joint-Gaussian model. The variable describing the signal component is defined as

$$\tilde{\boldsymbol{y}}_{kl} = \boldsymbol{U}_{kl(q)}^T(\boldsymbol{x} - \tilde{\boldsymbol{m}}_{kl}). \quad (6)$$

According to (5), $\tilde{\boldsymbol{y}}_{kl} \sim \mathcal{N}\left(\mathbf{o}, \boldsymbol{\Lambda}_{kl(q)}\right)$.

2) Probability Principal Components Analysis Model (PPCA Model): Subspace approximation model directly discards the noise subspace, while PPCA model concerns the statistical modeling for the noise subspace. The generative model of a d-dimensional HRRP sample in the lth $(l = 1, 2, \ldots, L_k)$ aspect-frame of the target $T_k (k = 1, 2, \ldots, K)$ is given by $\boldsymbol{x} = \boldsymbol{A}_{kl}\boldsymbol{y}_{kl} + \boldsymbol{\mu}_{kl} + \boldsymbol{\varepsilon}_{kl}$, where $\boldsymbol{\mu}_{kl} \in \Re^d$ is the mean vector, $\boldsymbol{A}_{kl} \in \Re^{d \times q}$ is the weight matrix with $\boldsymbol{a}_{klj_1}^T\boldsymbol{a}_{klj_2} = c(j_1, j_2 = 1, 2, \ldots q)$ in which c is a constant, and when $j_1 \neq j_2$, $c = 0$; otherwise, $c \neq 0$, i.e., $\boldsymbol{A}_{kl}^T\boldsymbol{A}_{kl}$ being a diagonal matrix, the latent variable $\boldsymbol{y}_{kl} \in \Re^q \sim \mathcal{N}(\mathbf{o}, \boldsymbol{I}_q)$ with $q < d$, and the noise variable $\boldsymbol{\varepsilon}_{kl} \in \Re^d \sim \mathcal{N}\left(\mathbf{o}, \sigma_{kl}^2\boldsymbol{I}_d\right)$, then $\boldsymbol{x} \sim \mathcal{N}\left(\boldsymbol{m}_{kl} = \boldsymbol{\mu}_{kl}, \boldsymbol{C}_{kl} = \sigma_{kl}^2\boldsymbol{I}_d + \boldsymbol{A}_{kl}\boldsymbol{A}_{kl}^T\right)$, namely

$$p(\boldsymbol{x} \mid l; k) = (2\pi)^{-d/2}\left|\sigma_{kl}^2\boldsymbol{I}_d + \boldsymbol{A}_{kl}\boldsymbol{A}_{kl}^T\right|^{-1/2} \times \exp\left\{-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_{kl})^T\left(\sigma_{kl}^2\boldsymbol{I}_d + \boldsymbol{A}_{kl}\boldsymbol{A}_{kl}^T\right)^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_{kl})\right\}. \quad (7)$$

Let $\boldsymbol{X}_{kl} = \left\{\boldsymbol{x}_{kli} \in \Re^d \mid i = 1, 2, \ldots, N\right\}$ denote the set of training data. The log-likelihood of the training dataset under the distribution model given by (7) is

$$\begin{aligned} L(\boldsymbol{\Theta}_{kl}) &= \sum_{i=1}^{N}\ln\{p(\boldsymbol{x}_{kli})\} \\ &= -\frac{N}{2}\left\{d\ln(2\pi) + \ln\left(\left(\sigma_{kl}^2\right)^d\left|\sigma_{kl}^{-2}\boldsymbol{M}_{kl}\right|\right) + \operatorname{tr}\left\{\sigma_{kl}^{-2}\left(\boldsymbol{I}_d - \boldsymbol{A}_{kl}\boldsymbol{M}_{kl}^{-1}\boldsymbol{A}_{kl}^T\right) \times \left[\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{x}_{kli} - \boldsymbol{\mu}_{kl})(\boldsymbol{x}_{kli} - \boldsymbol{\mu}_{kl})^T\right]\right\}\right\} \end{aligned} \quad (8)$$

where $\boldsymbol{\Theta}_{kl} = \left\{\boldsymbol{\mu}_{kl}, \sigma_{kl}^2, \boldsymbol{A}_{kl}\right\}$ denotes the unknown parameters in this model, and $\boldsymbol{M}_{kl} = \sigma_{kl}^2 \boldsymbol{I}_q + \boldsymbol{A}_{kl}^T \boldsymbol{A}_{kl}$. By formula derivation, $|\boldsymbol{C}_{kl}| = \left(\sigma_{kl}^2\right)^d \left|\sigma_{kl}^{-2}\boldsymbol{M}_{kl}\right|$ and $\boldsymbol{C}_{kl}^{-1} = \sigma_{kl}^{-2}\left(\boldsymbol{I}_d - \boldsymbol{A}_{kl}\boldsymbol{M}_{kl}^{-1}\boldsymbol{A}_{kl}^T\right)$. According to (8), the ML estimators of $\boldsymbol{\mu}_{kl}$, σ_{kl}^2 and $\boldsymbol{A}_{kl}$ are, respectively

$$\begin{aligned}\tilde{\boldsymbol{\mu}}_{kl} &= \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{x}_{kli} \\ \tilde{\sigma}_{kl}^2 &= \frac{1}{d-q}\sum_{j=q+1}^{d} \lambda_{klj} \\ \tilde{\boldsymbol{A}}_{kl} &= \boldsymbol{U}_{kl(q)}\left(\boldsymbol{\Lambda}_{kl(q)} - \tilde{\sigma}_{kl}^2 \boldsymbol{I}_q\right)^{1/2} \boldsymbol{R}\end{aligned} \tag{9}$$

where $\boldsymbol{U}_{kl(q)}$ and $\boldsymbol{\Lambda}_{kl(q)}$ are same as those in subspace approximation model, $\boldsymbol{R}$ is an arbitrary $q \times q$ orthogonal rotation matrix. Then

$$\begin{aligned}\tilde{\boldsymbol{C}}_{kl} &= \tilde{\sigma}_{kl}^2 \boldsymbol{I}_d + \tilde{\boldsymbol{A}}_{kl}\tilde{\boldsymbol{A}}_{kl}^T \\ &= \boldsymbol{U}_{kl(q)}\boldsymbol{\Lambda}_{kl(q)}\boldsymbol{U}_{kl(q)}^T + \left(\frac{1}{d-q}\sum_{j=q+1}^{d}\lambda_{klj}\right) \\ &\quad \times \boldsymbol{U}_{kl(d-q)}\boldsymbol{U}_{kl(d-q)}^T \\ &= \left[\boldsymbol{U}_{kl(q)} \quad \boldsymbol{U}_{kl(d-q)}\right] \begin{bmatrix} \boldsymbol{\Lambda}_{kl(q)} & \\ & \frac{1}{d-q}\sum\limits_{j=q+1}^{d}\lambda_{klj}\boldsymbol{I}_{d-q}\end{bmatrix} \\ &\quad \times \begin{bmatrix}\boldsymbol{U}_{kl(q)}^T \\ \boldsymbol{U}_{kl(d-q)}^T\end{bmatrix} \\ &= \boldsymbol{U}_{kl(d)}\boldsymbol{\Lambda}'_{kl(d)}\boldsymbol{U}_{kl(d)}^T.\end{aligned} \tag{10}$$

In PPCA model, the eigenvectors of the estimated covariance matrix $\tilde{\boldsymbol{C}}_{kl}$ is same as those of the sample covariance matrix $\boldsymbol{S}_{kl}$; while the $d-q$ small eigenvalues of $\tilde{\boldsymbol{C}}_{kl}$ are the mean of those of $\boldsymbol{S}_{kl}$, thereby $\tilde{\boldsymbol{C}}_{kl} \neq \boldsymbol{S}_{kl}$. Moreover, $\left|\tilde{\boldsymbol{C}}_{kl}\right| = \left|\begin{bmatrix}\boldsymbol{\Lambda}_{kl(q)} & \\ & \tilde{\sigma}_{kl}^2\boldsymbol{I}_{d-q}\end{bmatrix}\right|$ and $\tilde{\boldsymbol{C}}_{kl}^{-1} = \boldsymbol{U}_{kl(d)}\boldsymbol{\Lambda}_{kl(d)}^{\prime -1}\boldsymbol{U}_{kl(d)}^T$, then

$$\begin{aligned}p(\boldsymbol{x} \mid l;k) &= (2\pi)^{-d/2}\left|\begin{bmatrix}\boldsymbol{\Lambda}_{kl(q)} & \\ & \tilde{\sigma}_{kl}^2\boldsymbol{I}_{d-q}\end{bmatrix}\right|^{-1/2} \\ &\quad \times \exp\left\{-\frac{1}{2}\left(\boldsymbol{x} - \tilde{\boldsymbol{\mu}}_{kl}\right)^T \boldsymbol{U}_{kl(d)} \right. \\ &\quad \left. \times \begin{bmatrix}\boldsymbol{\Lambda}_{kl(q)}^{-1} & \\ & \tilde{\sigma}_{kl}^{-2}\boldsymbol{I}_{d-q}\end{bmatrix} \boldsymbol{U}_{kl(d)}^T\left(\boldsymbol{x} - \tilde{\boldsymbol{\mu}}_{kl}\right)\right\}.\end{aligned} \tag{11}$$

Comparing (5) with (11), we can see that subspace approximation model and PPCA model describe the statistical characteristics of the signal subspaces in the same way, while subspace model discards the noise subspace and PPCA model reserves some statistical characteristics of the noise subspace. Since

$$\begin{aligned}E\left(\boldsymbol{y}_{kl} \mid \boldsymbol{x}; \boldsymbol{\Theta}_{kl}\right) &= \left(\sigma_{kl}^2\boldsymbol{I}_q + \boldsymbol{A}_{kl}^T\boldsymbol{A}_{kl}\right)^{-1}\boldsymbol{A}_{kl}^T\left(\boldsymbol{x} - \boldsymbol{\mu}_{kl}\right) \\ &= \boldsymbol{M}_{kl}^{-1}\boldsymbol{A}_{kl}^T\left(\boldsymbol{x} - \boldsymbol{\mu}_{kl}\right)\end{aligned} \tag{12}$$

the q-dimensional latent variable and the d-dimensional noise variable of a d-dimensional HRRP sample $\boldsymbol{x}$ in the lth aspect-frame of the target T_k are

$$\begin{aligned}\tilde{\boldsymbol{y}}_{kl} &= \tilde{\boldsymbol{M}}_{kl}^{-1}\tilde{\boldsymbol{A}}_{kl}^T\left(\boldsymbol{x} - \tilde{\boldsymbol{\mu}}_{kl}\right) \\ &= \boldsymbol{\Lambda}_{kl(q)}^{-1}\boldsymbol{R}^T\left(\boldsymbol{\Lambda}_{kl(q)} - \tilde{\sigma}_{kl}^2\boldsymbol{I}_q\right)^{(1/2)}\boldsymbol{U}_{kl(q)}^T\left(\boldsymbol{x} - \tilde{\boldsymbol{\mu}}_{kl}\right) \\ \tilde{\boldsymbol{\varepsilon}}_{kl} &= \boldsymbol{x} - \tilde{\boldsymbol{A}}_{kl}\tilde{\boldsymbol{y}}_{kl} - \tilde{\boldsymbol{\mu}}_{kl}.\end{aligned} \tag{13}$$

In PPCA model, $\tilde{\boldsymbol{y}}_{kl} \sim \mathcal{N}(\mathbf{o}, \boldsymbol{I}_q)$ and $\tilde{\boldsymbol{\varepsilon}}_{kl} \sim \mathcal{N}\left(\mathbf{o}, \tilde{\sigma}_{kl}^2\boldsymbol{I}_d\right)$. If the variable defined by (6) is referred to as the latent variable in subspace approximation model, both of the latent variables in subspace approximation model and PPCA model follow the independent Gaussian distributions with zero mean vectors but different variance vectors. The variances of the elements in the latent variable in subspace approximation model are the corresponding eigenvalues, while those in PPCA model are normalized by not only the corresponding eigenvalues but also the variance of the noise variable. Thus, we can image that the statistical descriptions' exactnesses for the elements with larger eigenvalues in the two latent variables are similar, but the difference tends to be larger with the eigenvalues decreasing.

3) FA Model: PPCA model requires orthogonal bases in the signal subspace and independent elements in the noise variable with same variances, while FA model requires independent bases in the signal subspace and independent elements in the noise variable with different variances. The generative model of a d-dimensional HRRP sample in the lth $(l = 1, 2, \ldots, L_k)$ aspect-frame of the target $T_k(k = 1, 2, \ldots, K)$ is given by $\boldsymbol{x} = \boldsymbol{A}_{kl}\boldsymbol{y}_{kl} + \boldsymbol{\mu}_{kl} + \boldsymbol{\varepsilon}_{kl}$, where $\boldsymbol{\mu}_{kl} \in \Re^d$ is the mean vector, $\boldsymbol{A}_{kl} \in \Re^{d\times q}$ is the loading matrix with $\boldsymbol{A}_{kl}^T\boldsymbol{A}_{kl}$ being a non-diagonal matrix, the latent variable $\boldsymbol{y}_{kl} \in \Re^q \sim \mathcal{N}(\mathbf{o}, \boldsymbol{I}_q)$ with q denoting the number of factors and $q < d$, the noise variable $\boldsymbol{\varepsilon}_{kl} \in \Re^d \sim \mathcal{N}(\mathbf{o}, \boldsymbol{\Psi}_{kl})$ in which $\boldsymbol{\Psi}_{kl}$ is a diagonal matrix with the diagonal elements being different, then $\boldsymbol{x} \sim \mathcal{N}\left(\boldsymbol{m}_{kl} = \boldsymbol{\mu}_{kl}, \boldsymbol{C}_{kl} = \boldsymbol{\Psi}_{kl} + \boldsymbol{A}_{kl}\boldsymbol{A}_{kl}^T\right)$, namely

$$\begin{aligned}p(\boldsymbol{x} \mid l;k) =&\ (2\pi)^{-d/2}\left|\boldsymbol{\Psi}_{kl} + \boldsymbol{A}_{kl}\boldsymbol{A}_{kl}^T\right|^{-1/2} \\ &\times \exp\left\{-\frac{1}{2}\left(\boldsymbol{x} - \boldsymbol{\mu}_{kl}\right)^T\left(\boldsymbol{\Psi}_{kl} + \boldsymbol{A}_{kl}\boldsymbol{A}_{kl}^T\right)^{-1}\left(\boldsymbol{x} - \boldsymbol{\mu}_{kl}\right)\right\}.\end{aligned} \tag{14}$$

Let $\boldsymbol{X}_{kl} = \left\{\boldsymbol{x}_{kli} \in \Re^d \mid i = 1, 2, \ldots, N\right\}$ denote the set of training data. The log-likelihood of the training dataset under the distribution model given by (14) is

$$\begin{aligned}L(\boldsymbol{\Theta}_{kl}) &= \sum_{i=1}^{N}\ln\left\{p\left(\boldsymbol{x}_{kli}\right)\right\} \\ &= -\frac{N}{2}\left\{d\ln(2\pi) + \ln\left(|\boldsymbol{C}_{kl}|\right)\right. \\ &\quad + \operatorname{tr}\left\{\left(\boldsymbol{\Psi}_{kl}^{-1} - \boldsymbol{\Psi}_{kl}^{-1}\boldsymbol{A}_{kl}\boldsymbol{M}_{kl}^{-1}\boldsymbol{A}_{kl}^T\boldsymbol{\Psi}_{kl}^{-1}\right)\right. \\ &\quad \left.\left. \times \left[\frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{x}_{kli} - \boldsymbol{\mu}_{kl}\right)\left(\boldsymbol{x}_{kli} - \boldsymbol{\mu}_{kl}\right)^T\right]\right\}\right\}\end{aligned} \tag{15}$$

where $\boldsymbol{\Theta}_{kl}=\{\boldsymbol{\mu}_{kl},\boldsymbol{A}_{kl},\boldsymbol{\Psi}_{kl}\}$ denotes the unknown parameters in this model, and $\boldsymbol{M}_{kl}=\boldsymbol{I}_q+\boldsymbol{A}_{kl}^T\boldsymbol{\Psi}_{kl}^{-1}\boldsymbol{A}_{kl}$. By formula derivation, $\boldsymbol{C}_{kl}^{-1}=\boldsymbol{\Psi}_{kl}^{-1}-\boldsymbol{\Psi}_{kl}^{-1}\boldsymbol{A}_{kl}\boldsymbol{M}_{kl}^{-1}\boldsymbol{A}_{kl}^T\boldsymbol{\Psi}_{kl}^{-1}$. According to (15), the ML estimators of $\boldsymbol{\mu}_{kl}$ is

$$\tilde{\boldsymbol{\mu}}_{kl}=\frac{1}{N}\sum_{i=1}^{N}\boldsymbol{x}_{kli}. \tag{16}$$

As discussed in the literature [20], [21], [26], the ML estimators of the parameters $\boldsymbol{\Theta}'_{kl}=\{\boldsymbol{A}_{kl},\boldsymbol{\Psi}_{kl}\}$ can be obtained by an iterative expectation–maximization (EM) algorithm. The EM algorithm mainly consists of the following two steps.

E-step: Assume that $\boldsymbol{\Theta}'_{kl}(\tau)=\{\boldsymbol{A}_{kl}(\tau),\boldsymbol{\Psi}_{kl}(\tau)\}$ denotes the estimators in the τth iteration.

$$\begin{aligned}
&E\left(\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kl};\boldsymbol{\Theta}'_{kl}(\tau)\right)\\
&\quad=\boldsymbol{A}_{kl}^T(\tau)\left(\boldsymbol{\Psi}_{kl}(\tau)+\boldsymbol{A}_{kl}(\tau)\boldsymbol{A}_{kl}^T(\tau)\right)^{-1}\bar{\boldsymbol{x}}_{kl}\\
&\quad=\boldsymbol{A}_{kl}^T(\tau)\boldsymbol{C}_{kl}^{-1}(\tau)\bar{\boldsymbol{x}}_{kl}
\end{aligned}\tag{17}$$

$$\begin{aligned}
&E\left(\boldsymbol{y}_{kl}^T\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kl};\boldsymbol{\Theta}'_{kl}(\tau)\right)\\
&\quad=\boldsymbol{I}_q-\boldsymbol{A}_{kl}^T(\tau)\boldsymbol{C}_{kl}^{-1}(\tau)\boldsymbol{A}_{kl}(\tau)\\
&\qquad+E\left(\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kl};\boldsymbol{\Theta}'_{kl}(\tau)\right)E^T\left(\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kl};\boldsymbol{\Theta}'_{kl}(\tau)\right)
\end{aligned}\tag{18}$$

where $\bar{\boldsymbol{x}}_{kl}=\boldsymbol{x}-\tilde{\boldsymbol{\mu}}_{kl}$. The expected log likelihood is [see (19), shown at the bottom of the page].

M-step: Maximizing **(19)** with respect to $\boldsymbol{A}_{kl}$ and $\boldsymbol{\Psi}_{kl}$. The ML estimators in the $\tau+1$ th iteration are given by (20), shown at the bottom of the page, where $\mathrm{Diag}(\boldsymbol{B})$ denotes the operation that let the non-diagonal elements in the matrix $\boldsymbol{B}$ be zeroes.

Therefore, $\tilde{\boldsymbol{C}}_{kl}=\tilde{\boldsymbol{\Psi}}_{kl}+\tilde{\boldsymbol{A}}_{kl}\tilde{\boldsymbol{A}}_{kl}^T$. Moreover, $\left|\tilde{\boldsymbol{C}}_{kl}\right|=\left|\boldsymbol{\Sigma}_{kl(d)}\right|$ with $\boldsymbol{\Sigma}_{kl(d)}$ being the corresponding eigenvalue matrix of $\tilde{\boldsymbol{C}}_{kl}$. Then

$$\begin{aligned}
p(\boldsymbol{x}\mid l;k)=(2\pi)^{-d/2}\left|\boldsymbol{\Sigma}_{kl(d)}\right|^{-1/2}\exp\Bigg\{&-\frac{1}{2}(\boldsymbol{x}-\tilde{\boldsymbol{\mu}}_{kl})^T\\
&\times\left(\tilde{\boldsymbol{\Psi}}_{kl}^{-1}-\tilde{\boldsymbol{\Psi}}_{kl}^{-1}\tilde{\boldsymbol{A}}_{kl}\tilde{\boldsymbol{M}}_{kl}^{-1}\tilde{\boldsymbol{A}}_{kl}^T\tilde{\boldsymbol{\Psi}}_{kl}^{-1}\right)(\boldsymbol{x}-\tilde{\boldsymbol{\mu}}_{kl})\Bigg\}
\end{aligned}\tag{21}$$

where $\tilde{\boldsymbol{M}}_{kl}=\boldsymbol{I}_q+\tilde{\boldsymbol{A}}_{kl}^T\tilde{\boldsymbol{\Psi}}_{kl}^{-1}\tilde{\boldsymbol{A}}_{kl}$. According to (17), the q-dimensional latent variable and the d-dimensional noise variable of a d-dimensional HRRP sample $\boldsymbol{x}$ in the lth aspect-frame of the target T_k are

$$\begin{aligned}
\tilde{\boldsymbol{y}}_{kl}&=\tilde{\boldsymbol{A}}_{kl}^T\tilde{\boldsymbol{C}}_{kl}^{-1}(\boldsymbol{x}-\tilde{\boldsymbol{\mu}}_{kl})=\tilde{\boldsymbol{\Psi}}_{kl}^{-1}-\tilde{\boldsymbol{\Psi}}_{kl}^{-1}\tilde{\boldsymbol{A}}_{kl}\tilde{\boldsymbol{M}}_{kl}^{-1}\tilde{\boldsymbol{A}}_{kl}^T\tilde{\boldsymbol{\Psi}}_{kl}^{-1}\\
\tilde{\varepsilon}_{kl}&=\boldsymbol{x}-\tilde{\boldsymbol{A}}_{kl}\tilde{\boldsymbol{y}}_{kl}-\tilde{\boldsymbol{\mu}}_{kl}.
\end{aligned}\tag{22}$$

In the FA model, $\tilde{\boldsymbol{y}}_{kl}\sim\mathcal{N}(\mathbf{o},\boldsymbol{I}_q)$ and $\tilde{\varepsilon}_{kl}\sim\mathcal{N}\left(\mathbf{o},\tilde{\boldsymbol{\Psi}}_{kl}\right)$. Comparing (21) and (22) with (11) and (13), we can see that the statistical distribution and the latent variable derived by FA model are different from those by PPCA model despite of their generative models being similar.

C. Experimental Results Based on Measured Data

1) Data Description: The results presented in this paper are based on measured airplane data [9], [15]–[17]. The parameters of the targets and radar are shown in Table I, and the projections of target trajectories onto ground plane are shown in Fig. 1, from which the aspect angle of an airplane can be estimated according to its relative position to radar. As shown in Fig. 1, the measured data are segmented. In order to establish different frame models for each target and test the generalization performance of the recognition methods, there are two requirements for choosing

$$\begin{aligned}
Q\left(\boldsymbol{\Theta}'_{kl};\boldsymbol{\Theta}'_{kl}(\tau)\right)&=E_{\boldsymbol{y}|\bar{\boldsymbol{x}}}\left\{\sum_{i=1}^{N}\ln\left\{p\left(\bar{\boldsymbol{x}}_{kli}\mid\boldsymbol{y}_{kl};\boldsymbol{\Theta}'_{kl}(\tau)\right)\right\}\right\}\\
&=E_{\boldsymbol{y}|\bar{\boldsymbol{x}}}\left\{\sum_{i=1}^{N}\ln\left\{(2\pi)^{-d/2}\left|\boldsymbol{\Psi}_{kl}\right|^{-1/2}\exp\left\{-\frac{1}{2}(\bar{\boldsymbol{x}}_{kli}-\boldsymbol{A}_{kl}\boldsymbol{y}_{kl})^T\boldsymbol{\Psi}_{kl}^{-1}(\bar{\boldsymbol{x}}_{kli}-\boldsymbol{A}_{kl}\boldsymbol{y}_{kl})\right\}\right\}\right\}\\
&=-\frac{Nd}{2}\ln(2\pi)-\frac{N}{2}\ln\left(\left|\boldsymbol{\Psi}_{kl}\right|\right)\\
&\quad-\sum_{i=1}^{N}\left\{\begin{array}{l}\frac{1}{2}\bar{\boldsymbol{x}}_{kli}^T\boldsymbol{\Psi}_{kl}^{-1}\bar{\boldsymbol{x}}_{kli}-\bar{\boldsymbol{x}}_{kli}^T\boldsymbol{\Psi}_{kl}^{-1}\boldsymbol{A}_{kl}E\left(\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kli};\boldsymbol{\Theta}'_{kl}(\tau)\right)\\ +\frac{1}{2}tr\left[\boldsymbol{A}_{kl}^T\boldsymbol{\Psi}_{kl}^{-1}\boldsymbol{A}_{kl}E\left(\boldsymbol{y}_{kl}^T\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kli};\boldsymbol{\Theta}'_{kl}(\tau)\right)\right]\end{array}\right\}.
\end{aligned}\tag{19}$$

$$\begin{cases}
\tilde{\boldsymbol{A}}_{kl}(\tau+1)=\left[\sum\limits_{i=1}^{N}\boldsymbol{x}_{kli}E\left(\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kli};\boldsymbol{\Theta}'_{kl}\right)^T\right]\left[\sum\limits_{i=1}^{N}E\left(\boldsymbol{y}_{kl}\boldsymbol{y}_{kl}^T\mid\bar{\boldsymbol{x}}_{kli};\boldsymbol{\Theta}'_{kl}\right)\right]^{-1}\\
\tilde{\boldsymbol{\Psi}}_{kl}(\tau+1)=\frac{1}{N}\mathrm{Diag}\left\{\sum\limits_{i=1}^{N}\bar{\boldsymbol{x}}_{kli}\bar{\boldsymbol{x}}_{kli}^T-\tilde{\boldsymbol{A}}_{kl}(\tau+1)E\left(\boldsymbol{y}_{kl}\mid\bar{\boldsymbol{x}}_{kli};\boldsymbol{\Theta}'_{kl}\right)\bar{\boldsymbol{x}}_{kli}^T\right\}.
\end{cases}\tag{20}$$

TABLE I
PARAMETERS OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 *MHz*
	bandwidth		400 *MHz*
planes	length (*m*)	width (*m*)	height (*m*)
Yak-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

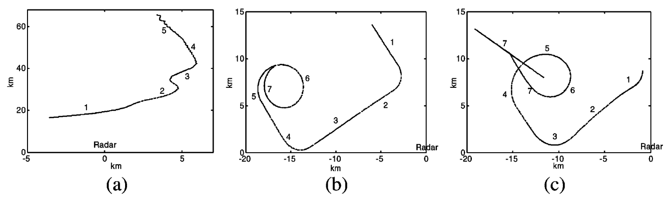

Fig. 1. Projections of target trajectories onto ground plane. (a) Yak-42; (b) Cessna Citation S/II; and (c) An-26.

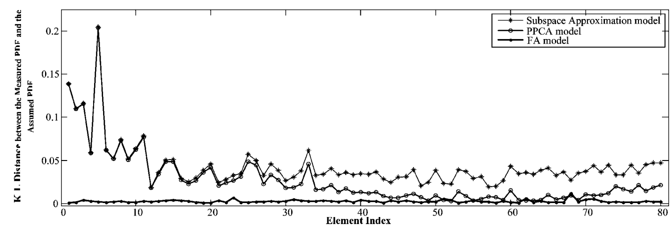

Fig. 2. KL distances between the measured pdf and the assumed pdf for all elements in the latent variables in subspace approximation model, PPCA model, and FA model from a measured HRRP frame of Yak-42.

the training data and test data: a) the training data cover almost all of the target-aspect angles of the test data and b) the elevation angles of the test data are different from those of the training data. Therefore, the second and the fifth segments of Yak-42, the sixth and the seventh segments of Cessna Citation S/ II, the fifth and the sixth segments of An-26 are taken as the training samples, other data segments are taken as test samples in our experiments. With large SNR, the effect of noise in the measured data on the recognition can be ignored. In addition, the measured HRRP is a 256-dimensional vector.

In the experiments in Section II-C, the aspect-frames in the training data are divided by the equal interval partition approach, and the latent variables in all aspect-frames have the same dimensionalities. There are 35 aspect-frames for Yak-42, 50 for Cessna Citation S/ II, and 50 for An-26, of which each frame has 1024 HRRP samples.

2) Statistical Descriptions' Exactness of the Three Joint-Gaussian Models: The three statistical models are proposed under the hypothesis that an HRRP frame follows a joint-Gaussian distribution. Whether the assumption is true or not will be discussed in the following.

As discussed in Section I, the KL distance between the measured pdf, which is estimated by the Parzen window method for measured HRRP samples, and the assumed pdf in a parametric model is used to evaluate the statistical description's exactnesses for the parametric model. The KL distance between $p_1(\boldsymbol{v})$ and $p_2(\boldsymbol{v})$ with $\boldsymbol{v} \in \Re^{d'}$ is defined as

$$\mathrm{KL}\left[p_1(\boldsymbol{v}) \| p_2(\boldsymbol{v})\right] \triangleq \int \cdots \int_{-\infty}^{+\infty} \left[p_1(\boldsymbol{v}) - p_2(\boldsymbol{v})\right] \ln \frac{p_1(\boldsymbol{v})}{p_2(\boldsymbol{v})} dv_1 \ldots dv_{d'}. \quad (23)$$

Obviously, $\mathrm{KL}\left[p_1(\boldsymbol{v}) \| p_2(\boldsymbol{v})\right] \geq 0$, the equal sign holds if and only if $p_1(\boldsymbol{v}) = p_2(\boldsymbol{v})$. In order to reduce the computation burden, we use the summation of the squared mean and the variance of the all KL distances between the measured pdfs and the assumed pdfs for all independent elements in a latent variable or a noise variable in a parametric model as the exactness criterion for the parametric model, and substitute the discrete summation operation within an finite region (usually, $[m - 3\sigma, m + 3\sigma]$ for Gaussian distribution, where m and σ respectively denote the mean and standard deviation of the element) for the integral operation.

According to Section II-B, the latent variables in the three parametric models follow independent Gaussian distributions, of which $\tilde{\boldsymbol{y}}_{kl|\text{S-A}} \sim \mathcal{N}\left(\mathbf{o}, \boldsymbol{\Lambda}_{kl(q)}\right)$, $\tilde{\boldsymbol{y}}_{kl|\text{PPCA}} \sim \mathcal{N}\left(\mathbf{o}, \boldsymbol{I}_q\right)$ and $\tilde{\boldsymbol{y}}_{kl|\text{FA}} \sim \mathcal{N}\left(\mathbf{o}, \boldsymbol{I}_q\right)$. Since comparing between the latent variables and noise variables in the three models, we take $q = 80$ in the following experiments. The KL distances for all elements in the latent variables in the three parametric models from a measured HRRP frame of Yak-42 are shown in Fig. 2, of which $m^2_{\mathrm{KL}_{\boldsymbol{y}}|\text{S-A}} + \sigma^2_{\mathrm{KL}_{\boldsymbol{y}}|\text{S-A}} = 2.6 \times 10^{-3}$, $m^2_{\mathrm{KL}_{\boldsymbol{y}}|\text{PPCA}} + \sigma^2_{\mathrm{KL}_{\boldsymbol{y}}|\text{PPCA}} = 1.8 \times 10^{-3}$ and $m^2_{\mathrm{KL}_{\boldsymbol{y}}|\text{FA}} + \sigma^2_{\mathrm{KL}_{\boldsymbol{y}}|\text{FA}} = 8.0 \times 10^{-6}$. Fig. 3 shows the six approximated pdfs from the HRRP frame same as that used in Fig. 2, every two of which correspond to the elements with the two largest KL distances in the latent variable in a parametric model, and where the bar represents the pdf approximated by histogram method, the dotted line represents that by Pazern windows using Gaussian kernels and the solid line represents that by the parametric model. Obviously, the assumed pdf for latent variable in FA model is very close to the measured pdf. Moreover, as shown in Figs. 2 and 3, the elements corresponding to larger eigenvalues in the latent variables in subspace approximation model and PPCA model depart from the assumed distribution forms more seriously than those corresponding to smaller eigenvalues, and the relation between the two latent variables accord with our analysis in Section II-B-2). Fig. 4 compares the statistical descriptions' exactnesses of the latent variables in the three parametric models for all aspect-frames in the training data by the criterion $m^2_{\mathrm{KL}_{\boldsymbol{y}}} + \sigma^2_{\mathrm{KL}_{\boldsymbol{y}}}$, which demonstrates that FA model can exactly describe the signal subspace in an HRRP frame.

In addition, the noise variables in PPCA and FA models also follow independent Gaussian distributions, of which $\tilde{\varepsilon}_{kl|\text{PPCA}} \sim \mathcal{N}\left(\mathbf{o}, \sigma^2_{kl} \boldsymbol{I}_d\right)$ and $\varepsilon_{kl|\text{FA}} \sim \mathcal{N}\left(\mathbf{o}, \boldsymbol{\Psi}_{kl}\right)$. Figs. 5 and 6 show the KL distances for all elements in the noise variables in the two models and the corresponding approximated pdfs from

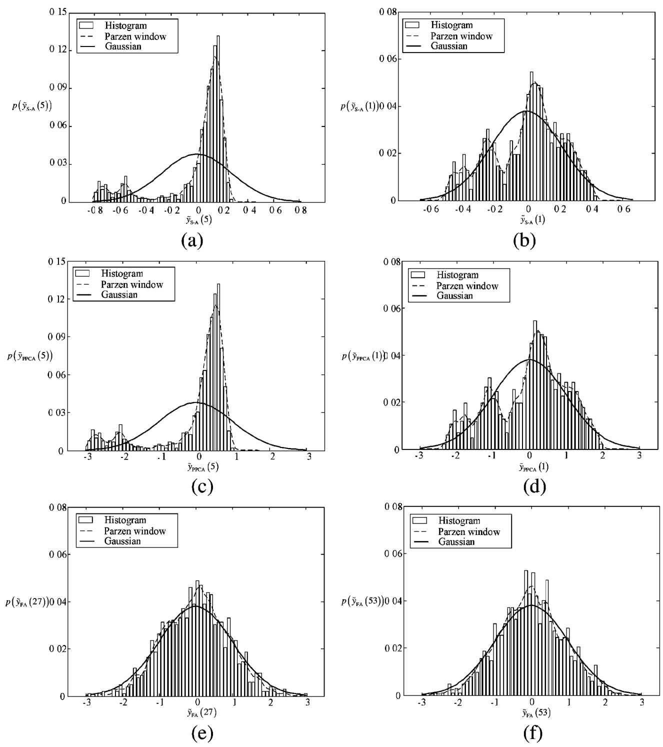

Fig. 3. Approximated pdfs of the elements with the two largest KL distances in the latent variables in subspace approximation model, PPCA model and FA model from the HRRP frame same as that used in Fig. 2. (a) $\tilde{y}_{\text{S-A}}(5)$ in subspace approximation model; (b) $\tilde{y}_{\text{S-A}}(1)$ in subspace approximation model; (c) $\tilde{y}_{\text{PPCA}}(5)$ in PPCA model; (d) $\tilde{y}_{\text{PPCA}}(1)$ in PPCA model; (e) $\tilde{y}_{\text{FA}}(27)$ in FA model; and (f) $\tilde{y}_{\text{FA}}(53)$ in FA model.

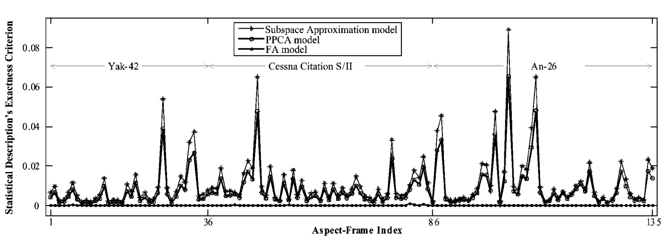

Fig. 4. Statistical description's exactnesses of the latent variables in subspace approximation model, PPCA model, and FA model for all aspect-frames in the training data by the criterion $m_{\text{KL}_y}^2 + \sigma_{\text{KL}_y}^2$.

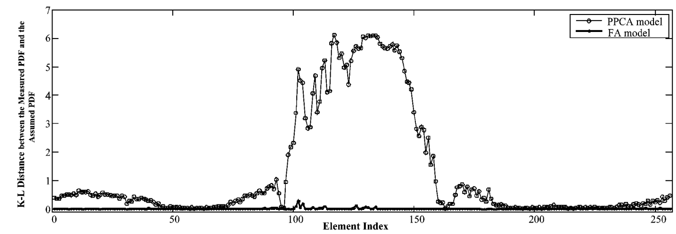

Fig. 5. KL distances between the measured pdf and the assumed pdf for all elements in the noise variables in PPCA model and FA model from the HRRP frame same as that used in Figs. 2 and 3.

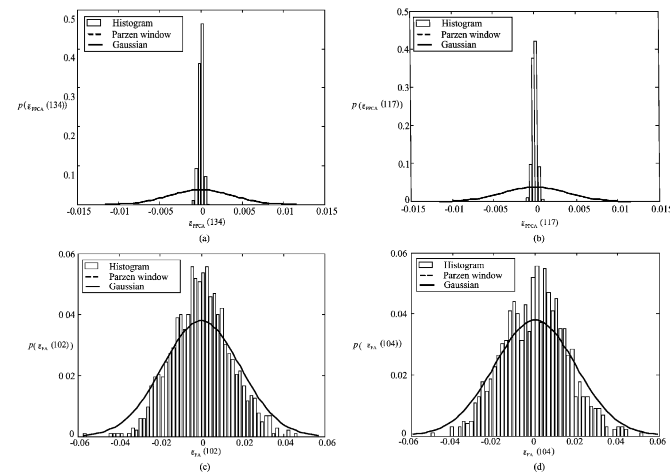

Fig. 6. Approximated pdfs of the elements with the two largest KL distances in the noise variables in PPCA model and FA model from the HRRP frame same as that used in Figs. 2, 3 and 5. (a) $\tilde{\varepsilon}_{\text{PPCA}}(134)$ in PPCA model, (b) $\tilde{\varepsilon}_{\text{PPCA}}(117)$ in PPCA model, (c) $\tilde{\varepsilon}_{\text{FA}}(102)$ in FA model (d) $\tilde{\varepsilon}_{\text{FA}}(104)$ in FA model.

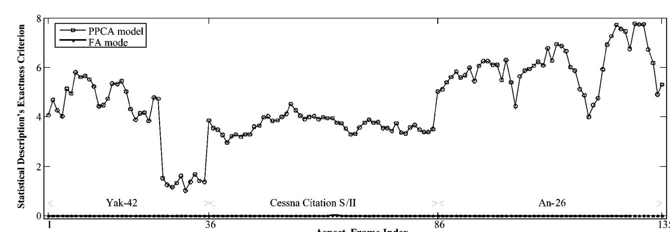

Fig. 7. Statistical description's exactnesses of the noise variables in PPCA model and FA model for all aspect-frames in the training data by the criterion $m_{\text{KL}_\varepsilon}^2 + \sigma_{\text{KL}_\varepsilon}^2$.

the HRRP frame same as that used in Figs. 2 and 3. As shown in Fig. 5, the statistical description's exactnesses for the part of an HRRP sample including target signal, i.e., the elements within the support region, in PPCA model are much lower than those in FA model. Furthermore, as shown in Fig. 6(a) and (b), the elements in this part in PPCA model approximate to follow Gaussian distribution, but the assumed variances largely depart from the measured variances, which causes the large KL distances. Here $m_{KL_{\boldsymbol{\mathcal{E}}}|\text{PPCA}}^2 + \sigma_{\text{KL}_{\boldsymbol{\mathcal{E}}}|\text{PPCA}}^2 = 5.5$ and $m_{\text{KL}_{\boldsymbol{\mathcal{E}}}|\text{FA}}^2 + \sigma_{\text{KL}_{\boldsymbol{\mathcal{E}}}|\text{FA}}^2 = 8.3 \times 10^{-4}$. Therefore, the assumed pdf for noise variable in FA model is very close to the measured pdf. Fig. 7 compares the statistical descriptions' exactnesses of the noise variables in PPCA and FA models for all aspect-frames in the training data by the criterion $m_{\text{KL}_\varepsilon}^2 + \sigma_{\text{KL}_\varepsilon}^2$, which demonstrates that FA model can exactly describe the noise subspace in an HRRP frame.

3) Recognition Experiments: The average recognition rates of subspace approximation model, PPCA model and FA model versus the dimensionality of the latent variable q are shown in Fig. 8, of which $q_{\text{S-A}}^* = 160$, $q_{\text{PPCA}}^* = 120$ and $q_{\text{FA}}^* = 20$. For the three models, the recognition rate of subspace approximation model which discards the noise subspace does not straightly increase with q, i.e., $q_{\text{S-A}}^* < d$; PPCA model which utilizes some statistical characteristics in the noise subspace largely improves the recognition performances; FA model whose requirements for the signal component and noise component are much looser achieves the best recognition result. Tables II and III show the confusion matrices and average recognition rates of subspace approximation model with $q_{\text{S-A}}^* = 160$, PPCA model with $q_{\text{PPCA}}^* = 120$, FA model with $q_{\text{FA}}^* = 20$, independent Gaussian model [6], independent Gamma model

TABLE II
CONFUSION MATRICES AND AVERAGE RECOGNITION RATES OF SUBSPACE APPROXIMATION MODEL WITH $q^*_{\text{S-A}} = 160$, PPCA MODEL WITH $q^*_{\text{PPCA}} = 120$, FA MODEL WITH $q^*_{\text{FA}} = 20$

	subspace approximation model ($q^* = 160$)			PPCA model ($q^* = 120$)			FA model ($q^* = 20$)		
	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26
Yak-42	99.83	1.60	21.00	100	4.90	14.50	100	1.90	6.85
Cessna Citation S/II	0	79.60	2.40	0	93.20	1.35	0	95.75	0.55
An-26	0.17	18.80	76.60	0	1.90	84.15	0	2.35	92.60
average recognition rates (%)	85.34			92.45			96.12		

TABLE III
CONFUSION MATRICES AND AVERAGE RECOGNITION RATES OF INDEPENDENT GAUSSIAN MODEL, INDEPENDENT GAMMA MODEL, INDEPENDENT TWO-DISTRIBUTION COMPOUNDED STATISTICAL MODEL

	independent Gaussian model			independent Gamma model			independent two-distribution compounded model		
	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26	Yak-42	Cessna Citation S/II	An-26
Yak-42	99.00	0.70	15.35	99.25	0.70	17.90	99.75	0	8.00
Cessna Citation S/II	0	89.50	9.55	0	89.50	1.40	0	93.60	1.60
An-26	1.00	9.80	75.10	0.75	9.80	80.70	0.25	6.40	90.40
average recognition rates (%)	87.87			89.82			94.58		

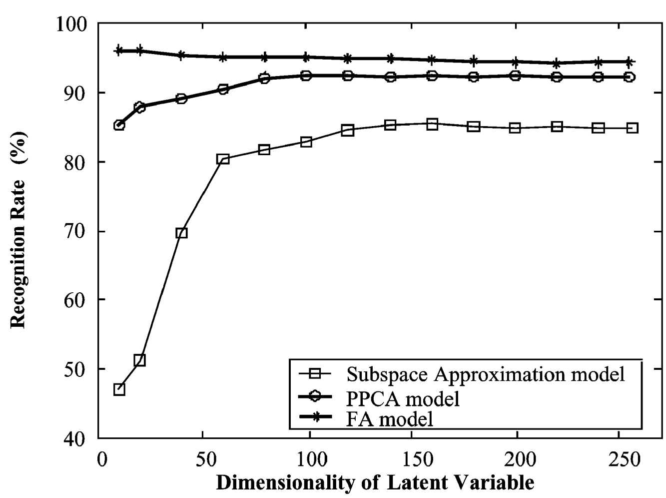

Fig. 8. Average recognition rates of subspace approximation model, PPCA model, and FA model versus the dimensionality of the latent variable.

[13], [14], independent two-distribution compounded statistical model [15]. In the six parametric models, FA model is the best one, independent two-distribution compounded statistical model is the next one, the average recognition rate of which is approximately 2 percentage points larger than that of PPCA model, and subspace approximation model is the last one, the average recognition rate of which is approximately 2.5 percentage points smaller than that of independent Gaussian model. This shows that the parametric models which consider the statistical correlation between the elements in an HRRP sample but cannot describe the jointly statistical characteristics accurately also cannot achieve good recognition performance.

4) *Discussion:* First, we will make a detailed comparison between the three joint-Gaussian models as follows.

a) *The latent variables are different:* The three parametric models all require that the elements in the latent variables follow independent Gaussian distribution. The orthogonal bases in signal subspace ensure the independence in subspace approximation model and PPCA model, while the EM algorithm ensures the independence in FA model. The orthogonal Gaussian variables must be independent, while the independent Gaussian variables may not be orthogonal. Therefore, the requirement for the latent variable in FA model is much looser.

b) *The statistical descriptions for the noise component are different:* Actually, only a part of an HRRP sample includes target signal, i.e., the support region, and the other mainly includes noises. Although noises are uniformly distributed in a returned echo vector along the range cells, an HRRP sample is obtained after the demodulation, during which the target signal can suppress the noises in the same range cells; therefore, the noises outside the support region are larger than those inside the support region for an HRRP sample. For the three models, in subspace approximation model, if $q < d$, the noise subspace is discarded, and if $q = d$, the full covariance is utilized, in which the noise component may deteriorate the recognition performance; in PPCA model and FA model, the statistical characteristics of the noise component are both utilized, but the noise variances inside and outside the support region are assumed to be equal in PPCA model, which is not true for radar HRRP samples according to our aforementioned analysis. Therefore, as shown in Section II-C-3), the recognition rates of subspace approximation model for $q = d$ are lower than that for $q < d$, the recognition performances of PPCA model and FA model are much better than those of subspace approximation model, and the statistical descriptions' exactnesses inside the support region in PPCA model are much lower than those in FA model.

c) *The computation burdens of parameter estimations are different:* The parameter estimations in subspace approximation model and PPCA model both depend on the estimation of the $d \times d$-dimensional sample variance matrix for each HRRP frame. Usually, an HRRP sample includes many range cells ($d = 256$ for the measured data in this paper), but the samples in an HRRP frame are not too many ($N = 1024$ in the equal interval partition approach in this paper). Thus, the computation burdens of parameter estimations for the two models are very large and their estimation exactnesses cannot be ensured. FA model does not estimate the sample variance matrix for each HRRP frame, but directly estimates the corresponding loading matrix and the variance matrix of the noise subspace by the EM algorithm. Therefore, the computation burden of parameter estimation in FA model decreases and its estimation exactness increases.

d) *The statistical descriptions' error distributions for non-Gaussian distributed correlations in HRRP data are different:* For a joint-Gaussian variable, all of its elements should follow Gaussian distributions. Nevertheless, according to the scattering center model, elements in HRRP samples are classified into three statistical types [15], [18], of which the first type follows a Rayleigh distribution, the second follows a Ricean distribution, and the third follows a multimodal (mainly bimodal) distribution. The experimental results in the literature [15] based on measured data also demonstrated that the elements in HRRP samples depart from Gaussian distribution. Thus, HRRP samples cannot strictly follow a joint-Gaussian distribution. Actually, we utilized the joint-Gaussian distributions to approximate the non-Gaussian distributed correlations in HRRP data in the aforementioned experiments. As shown in Section II-C-2), the elements corresponding to larger eigenvalues in the latent variables in subspace approximation model and PPCA model, which are the principal components in the signal subspace and can have main effect on the recognition results, depart from the assumed distribution forms more seriously than those corresponding to smaller eigenvalues, but all of the elements in the latent variable in FA model nearly approximate to the assumed distribution forms and the statistical descriptions' error in the signal subspace of FA model is smaller than that in the noise subspace. Therefore, the signal subspace in FA model for the non-Gaussian distributed correlations in HRRP data can nearly approximate to the independent-Gaussian distribution, which is helpful for the statistical recognition.

Second, without any prior information, a joint-Gaussian mixture model [19], [21], [25] is a choice for modeling non-Gaussian distributed correlations in multidimensional data. However, there are several problems for a joint-Gaussian mixture model, e.g., PPCA mixture model [19] or FA mixture model [21], applying to radar HRRP statistical recognition.

a) The recognition performance of a joint-Gaussian mixture model bears on the number of mixture components, but it is difficult to determine the optimal number of mixture components [19], [21], [25]. Moreover, the optimal dimensionality of the latent variable in each mixture component [19], [21], [27] and the optimal aspect-frame partition should also be determined.

b) Since an HRRP sample includes many range cells ($d = 256$ for the measured data in this paper), the parameter learning of a joint-Gaussian mixture model for HRRP samples is very difficult [19], [21], [25].

c) For a joint-Gaussian mixture model, its computation burden and storage requirement greatly increase not only in the training phase but also in the test phase, therefore, the recognition speed decreases.

Due to experimental results in Sections II-C-2) and II-C-3), we can have the conclusion that an HRRP frame approximately follows the joint-Gaussian distribution described by FA model. Even if the statistical description for an HRRP frame by a joint-Gaussian mixture model is more exact than that by FA model, its learning complexity is higher, its computation burden is larger, its recognition speed is lower, and it may not ensure good recognition rate, since too exact model may deteriorate the generalization performance. Therefore, we can apply FA model rather than a joint-Gaussian mixture model to radar HRRP statistical recognition, and the difficulty in statistical modeling for HRRP samples is largely reduced.

III. Model Selection for FA Model in Radar HRRP Statistical Recognition

A. Optimal Number of Factors

The problem that model selection addresses is the following: Given a set of observed data $\boldsymbol{X} = \left\{\boldsymbol{x}_i \in \Re^d \mid i = 1, 2, \ldots, N\right\}$ and a set of candidate models $\left\{\mathcal{M}_j : g_j\left(\boldsymbol{x}, \boldsymbol{\theta}_j\right)\right\}_{j=1}^{M}$, choose the best model that describes the observed data $\boldsymbol{X}$, where $g_j\left(\boldsymbol{x}, \boldsymbol{\theta}_j\right)$ is a vector function that represents the j th model, and $\boldsymbol{\theta}_j \in \Re^{m_j}$ is a vector of model parameters with m_j denoting the length of the vector, which can be estimated from the observed data $\boldsymbol{X}$ and its estimator is denoted as $\tilde{\boldsymbol{\theta}}_j$. The two most popular model selection criteria have been the AIC [23], [24] and BIC [24].

1) Akaike Information Criterion: The log likelihood ratio test statistic asymptotically satisfies the relation

$$\log p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right) - \log p\left(\boldsymbol{X} \mid \boldsymbol{\theta}_j, \mathcal{M}_j\right) = \chi^2\left(m_j\right) \quad (24)$$

where $\log p\left(\boldsymbol{X} \mid \boldsymbol{\theta}_j, \mathcal{M}_j\right)$ and $\log p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right)$ respectively denote the log-likelihoods of the observed data under the "true" distribution model and the estimated distribution model, $\chi^2\left(m_j\right)$ denotes a chi-squared variable with the degree of freedom m_j which is equal to the length of the parameter vector. By formula derivation

$$E\left[\log p\left(\boldsymbol{X} \mid \boldsymbol{\theta}_j, \mathcal{M}_j\right)\right] = E\left[\log p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right) - \chi^2\left(m_j\right)\right] = \log p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right) - m_j \quad (25)$$

where $E(\bullet)$ denotes the expectation with respect to $\boldsymbol{\theta}_j$. Thus, the AIC is

$$\begin{aligned} j^* &= \arg \max_{j=0,1,\ldots,M-1} \left\{E\left[\log p\left(\boldsymbol{X} \mid \boldsymbol{\theta}_j, \mathcal{M}_j\right)\right]\right\} \\ &= \arg \max_{j=0,1,\ldots,M-1} \left\{\log p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right) - m_j\right\} \\ &= \arg \max_{j=0,1,\ldots,M-1} \mathrm{AIC}(j). \end{aligned} \quad (26)$$

2) Bayesian Information Criterion: Under certain regularity conditions and large data samples, we can use Laplace's method for integration, then

$$\begin{aligned} p\left(\boldsymbol{X} \mid \mathcal{M}_j\right) &= \int_{\boldsymbol{\Theta}_j} p\left(\boldsymbol{X} \mid \boldsymbol{\theta}_j, \mathcal{M}_j\right) p\left(\boldsymbol{\theta}_j \mid \mathcal{M}_j\right) d\boldsymbol{\theta}_j \\ &\approx (2\pi)^{(m_j/2)} \left|\tilde{\boldsymbol{H}}_j\right|^{-(1/2)} \\ &\quad \times p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right) p\left(\tilde{\boldsymbol{\theta}}_j \mid \mathcal{M}_j\right) \end{aligned} \quad (27)$$

where $\boldsymbol{\Theta}_j$ is the model's parameter space, and $\tilde{\boldsymbol{H}}_j$ is the Hessian of $-\log p\left(\boldsymbol{X} \mid \boldsymbol{\theta}_j, \mathcal{M}_j\right)$ evaluated at $\boldsymbol{\theta} = \tilde{\boldsymbol{\theta}}$. When observed data are independent and identically distributed, we can write

$$\left|\tilde{\boldsymbol{H}}_j\right| = O\left(N^{(m_j/2)}\right). \quad (28)$$

Under the hypothesis that all of the candidate models have equal prior probabilities, the BIC is

$$\begin{aligned} j^* &= \arg \max_{j=0,1,\ldots,M-1} \left\{\log p\left(\boldsymbol{X} \mid \mathcal{M}_j\right)\right\} \\ &\approx \arg \max_{j=0,1,\ldots,M-1} \left\{\log p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right) - \frac{1}{2}\log\left|\tilde{\boldsymbol{H}}_j\right|\right\} \\ &= \arg \max_{j=0,1,\ldots,M-1} \left\{\log p\left(\boldsymbol{X} \mid \tilde{\boldsymbol{\theta}}_j, \mathcal{M}_j\right) - \frac{m_j}{2}\log N\right\} \\ &= \arg \max_{j=0,1,\ldots,M-1} \mathrm{BIC}(j). \end{aligned} \quad (29)$$

Comparing between (26) and (29), we can see that the AIC and BIC both consist of two terms: a data term which evaluates the parameter estimation and a penalty term which penalizes for overfitting. As the model complexity increases, the data term usually decreases, whereas the penalty term always increases. The best model is the one that yields the maximum value of the criterion. Since, for the number of observed samples N greater than 8, $\log N$ exceeds 2, the BIC favors models with fewer parameters than does the AIC.

Usually, model selection of FA model deals with how to automatically determine the optimal number of factors q. Let $g_j\left(\boldsymbol{x}, \boldsymbol{\theta}_j\right) : \boldsymbol{x} \in \Re^d \sim \mathcal{N}\left(\boldsymbol{\mu}_{j(d\times 1)}, \boldsymbol{\Psi}_{j(d\times d)} + \boldsymbol{A}_{j(d\times j)}\boldsymbol{A}_{j(d\times j)}^T\right)$ denote the jth $(j = 1, 2, \ldots, d-1)$ candidate model, in which the number of factors $q = j$. Then the length of the parameter vector is [22], [23]

$$m_j = d(j+1) - \frac{j(j-1)}{2}. \quad (30)$$

Substituting (30) into (26), the number of factors can be determined by the AIC. Similarly, substituting (30) into (29), the number of factors can be determined by the BIC.

B. Adaptive Aspect-Frame Partition Approach

Due to the target-aspect sensitivity of HRRP samples, how to divide aspect-frames for the training data is another important issue in model selection for radar HRRP statistical recognition. Since the scattering center model almost remains unchanged within a target-aspect sector without the scatterers' MTRC, the literature [2], [3], [9], [12], [15], [16] determined the number of aspect-frames in the whole training data according to a target-aspect sector without the scatterers' MTRC and divided the training data into target-aspect frames at equal interval, which is referred to as the equal interval partition approach in this paper. The experiments in Section II also adopt this approach. As we known, it is a prerequisite for ISAR imaging, not for radar HRRP statistical recognition, that the target-aspect change corresponding to an aspect-frame must avoid scatterers' MTRC. Thus, from the viewpoint of statistical recognition, the equal interval partition approach is not optimal.

Different from what discussed in Section III-A, the observed dataset is a parameter for model selection of FA model in radar HRRP statistical recognition, i.e., the candidate model are $\{\mathcal{M}'_{t,j} : g_{t,j}(\boldsymbol{x}, \boldsymbol{\theta}_j, \boldsymbol{X}_t = \{\boldsymbol{x}_i\}_{i=1}^{N_t})\}_{t=1,j=1}^{T,d-1}$, where $g_{t,j}(\boldsymbol{x}, \boldsymbol{\theta}_j, \boldsymbol{X}_t) : \boldsymbol{x} \sim \mathcal{N}(\boldsymbol{\mu}_{t,j(d\times 1)}, \boldsymbol{\Psi}_{t,j(d\times d)} + \boldsymbol{A}_{t,j(d\times j)}\boldsymbol{A}_{t,j(d\times j)}^T)$. The main idea of our approach to find the optimal partition approach consists of the two steps in the following, thus is referred to "two-step method" in this paper.

1) *Determining the optimal number of factors for each observed dataset:* For an observed dataset $\boldsymbol{X}_t = \{\boldsymbol{x}_i\}_{i=1}^{N_t}$,

$$\mathrm{AIC}(j \mid t) = \log p\left(\boldsymbol{X}_t \mid \tilde{\boldsymbol{\theta}}_j, M_j\right) - d(j+1) + \frac{j(j-1)}{2}, \quad j = 1, 2, \ldots, d-1 \quad (31)$$

$$\mathrm{BIC}(j \mid t) = \log p\left(\boldsymbol{X}_t \mid \tilde{\boldsymbol{\theta}}_j, M_j\right) - \left[\frac{d(j+1)}{2} - \frac{j(j-1)}{4}\right]\log N_t, \quad j = 1, 2, \ldots, d-1 \quad (32)$$

where $\tilde{\boldsymbol{\theta}}_j = \left\{\tilde{\boldsymbol{\mu}}_{j(d\times 1)}, \tilde{\boldsymbol{\Psi}}_{j(d\times d)}, \tilde{\boldsymbol{A}}_{j(d\times j)}\right\}$ is the ML estimators from $\boldsymbol{X}_t$. Substituting (31) into (26), or substituting (32) into (29), the optimal number of factors for each observed dataset j_t $(t = 1, 2, \ldots, T)$ can be determined.

2) *Comparing between the models with different observed datasets and the corresponding optimal numbers of factors:* Obviously, since $\boldsymbol{X}_t$ is a parameter here, it is not suitable to still take the criterion $\mathrm{AIC}(t) = E\left[\log p\left(\boldsymbol{X}_t \mid \boldsymbol{\theta}_{j_t}, \mathcal{M}_{j_t}\right)\right]$ or $\mathrm{BIC}(t) = \log p\left(\boldsymbol{X}_t \mid \mathcal{M}_{j_t}\right)$

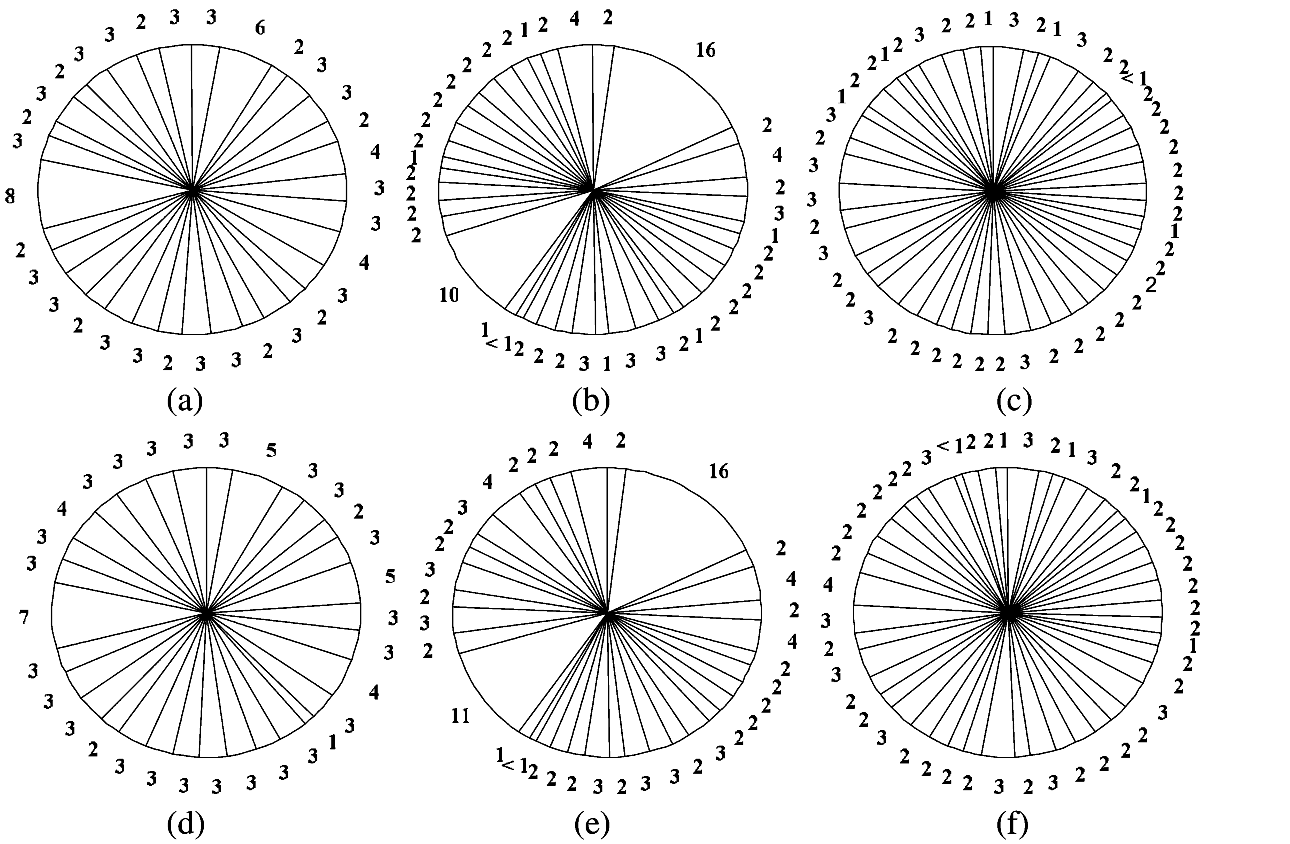

Fig. 9. Proportions of the number of samples in each aspect-frame to the number of all training samples by the adaptive partition approach based on the AIC and BIC with the frame step $\Delta N = 4(\%)$. (a) AIC-based partition for Yak-42 ($L_Y = 32$). (b) AIC-based partition for Cessna Citation S/II ($L_C = 39$). (c) AIC-based partition for An-26 ($L_A = 48$). (d) BIC-based partition for Yak-42 ($L_Y = 31$). (e) BIC-based partition for Cessna Citation S/II ($L_C = 35$). (f) BIC-based partition for An-26 ($L_A = 47$).

with $t = 1, 2, \ldots, T$. Considering that the model complexity increases with the number of observed samples, we take the criterion

$$\begin{aligned}\text{AIC}: \quad t^* &= \arg \max_{t=1,2,\ldots,T} M\text{AIC}(t)\\ &= \arg \max_{t=1,2,\ldots,T} E\left[\frac{\log p(\boldsymbol{X}_t \mid \boldsymbol{\theta}_{j_t}, \mathcal{M}_{j_t})}{N_t}\right]\\ &= \arg \max_{t=1,2,\ldots,T} \left[\frac{(AICj_t)}{N_t}\right] \qquad (33)\\ \text{BIC}: \quad t^* &= \arg \max_{t=1,2,\ldots,T} M\text{BIC}(t)\\ &= \arg \max_{t=1,2,\ldots,T} \left[\frac{\log p(\boldsymbol{X}_t \mid \mathcal{M}_{j_t})}{N_t}\right]\\ &= \arg \max_{t=1,2,\ldots,T} \left[\frac{\text{BIC}(j_t)}{N_t}\right] \qquad (34)\end{aligned}$$

where the optimal model is $\mathcal{M}'_{t^*,j_{t^*}} : g_{t^*,j_{t^*}}(\boldsymbol{x}, \boldsymbol{\theta}_{j_{t^*}}, \boldsymbol{X}_{t^*})$. In other words, the larger the average AIC coefficient or the average BIC coefficient is, the better the model with the observed dataset as a parameter is.

Since the training data are measured continuously, the aspect-frame partition only concerns the determination of each frame length. Assume that the training data cover the whole aspect region (0°–360°), all samples are amplitude normalized and time-shift compensated within the corresponding aspect frames. The main procedure of the adaptive partition approach for the training data of target T_k is as follows.

Step 1) Initial parameters: Let $l = 1, m = 0$, Flag $= 0$, and S_l denote the starting point, and $S_1 = 0$. Compute the initial aspect-frame length

$$N^{(0)} = \left\lfloor \frac{N_{\text{all}}}{L_k} \right\rfloor = \left\lfloor N_{\text{all}} \cdot \frac{(\delta\varphi)_{\text{MTRC}}}{2\pi} \right\rfloor = \left\lfloor N_{\text{all}} \cdot \frac{C}{4\pi \cdot BL_x} \right\rfloor \qquad (35)$$

where $\lfloor \bullet \rfloor$ denotes rounding toward zeroes, and N_{all} denotes the number of all training samples.

Step 2) Initial partition and selection: If $S_l + N^{(0)} \leq N_{\text{all}}$, the initial lth frame is $\boldsymbol{X}_{kl}^{(0)} = \left\{\boldsymbol{x}_{kli}^{(0)} \in \Re^d \mid i = S_l + 1, S_l + 2, \ldots, S_l + N^{(0)}\right\}$; otherwise, the algorithm ends. Let ΔN denote the frame step ($\Delta N \ll N^{(0)}$). The initial candidate models are

$$\left\{\mathcal{M}'_{t,j} : g_{t,j}\left(\boldsymbol{x}, \boldsymbol{\theta}_j, \boldsymbol{X}_{klt}^{(0)} = \left\{\boldsymbol{x}_{kli}^{(0)}\right\}_{i=S_l+1}^{S_l+N_t}\right)\right\}_{t=1,j=1}^{T=3,d-1}$$

where $N_1 = N^{(0)}$, $N_2 = N^{(0)} - \Delta N$ and $N_3 = N^{(0)} + \Delta N$. Use the "two-step method" to select the optimal frame model $\mathcal{M}'^{(0)}_{t^*,j_{t^*}} : g_{t^*,j_{t^*}}\left(\boldsymbol{x}, \boldsymbol{\theta}_{j_{t^*}}, \boldsymbol{X}_{klt^*}^{(0)}\right)$.

Step 3) Initial judgement: If $t^* = 1$, let Flag $= 0$; if $t^* = 2$ and $N^{(0)} - 2\Delta N \geq d$, let Flag $= -1$; if $t^* = 2$ and $N^{(0)} -$

TABLE IV
CONFUSION MATRICES AND AVERAGE RECOGNITION RATES OBTAINED BY THE AIC AND BIC-BASED EQUAL INTERVAL PARTITION APPROACH AND THE AIC- AND BIC-BASED ADAPTIVE PARTITION APPROACH WITH $\Delta N = 4$

	equal interval partition approach						adaptive partition approach with $\Delta N = 4$					
	AIC ($L_Y = 35$, $L_C = 50$, $L_A = 50$)			BIC ($L_Y = 35$, $L_C = 50$, $L_A = 50$)			AIC ($L_Y = 32$, $L_C = 39$, $L_A = 48$)			BIC ($L_Y = 31$, $L_C = 35$, $L_A = 47$)		
	Yak -42	Cessna Citation S/II	An-2 6	Yak -42	Cessna Citation S/II	An-2 6	Yak -42	Cessna Citatio n S/II	An-2 6	Yak -42	Cessna Citation S/II	An-2 6
Yak-42	100	1.40	10.60	100	1.65	5.35	100	3.75	8.10	100	0.10	2.45
Cessna Citation S/II	0	95.15	1.05	0	96.25	0.55	0	96.05	1.35	0	99.00	0.35
An-26	0	3.45	88.35	0	2.10	94.10	0	0.20	90.55	0	0.90	97.20
average recog -nition rates (%)	94.50			96.78			95.53			98.73		

$2\Delta N < d$, let Flag $= 0$; if $t^* = 3$ and $N^{(0)} + 2\Delta N \le N_{\text{all}}$, let Flag $= 1$; if $t^* = 3$ and $N^{(0)} + 2\Delta N > N_{\text{all}}$, let Flag $= 2$.

Step 4) Iterative partition and selection: If Flag $= 0$, let $M_{kl}^{\prime *} : g_{t^*,j_{t^*}}\left(\boldsymbol{x}, \boldsymbol{\theta}_{j_{t^*}}, \boldsymbol{X}_{klt^*}^{(m)}\right)$, $S_{l+1} = S_l + N^{(m)}$, $l = l + 1$, and return to Step 2); if Flag $= -1$, let $m = m + 1$, $N^{(m)} = N^{(m)} - \Delta N$, the optimal frame model $\mathcal{M}_{t^*,j_{t^*}}^{\prime(m)} : g_{t^*,j_{t^*}}\left(\boldsymbol{x}, \boldsymbol{\theta}_{j_{t^*}}, \boldsymbol{X}_{klt^*}^{(m)}\right)$ is selected from the candidate models

$$\left\{\mathcal{M}_{t,j}' : g_{t,j}\left(\boldsymbol{x}, \boldsymbol{\theta}_j, \boldsymbol{X}_{klt}^{(m)} = \left\{\boldsymbol{x}_{kli}^{(m)}\right\}_{i=S_l+1}^{S_l+N_t}\right)\right\}_{t=1,j=1}^{T=2,d-1}$$

with $N_1 = N^{(m)}$ and $N_2 = N^{(m)} - \Delta N$ by the "two-step method", and return to Step 5); if Flag $= 1$, let $m = m + 1$, $N^{(m)} = N^{(m)} + \Delta N$, the optimal frame model $\mathcal{M}_{t^*,j_{t^*}}^{\prime(m)} : g_{t^*,j_{t^*}}\left(\boldsymbol{x}, \boldsymbol{\theta}_{j_{t^*}}, \boldsymbol{X}_{klt^*}^{(m)}\right)$ is selected from the candidate models

$$\left\{\mathcal{M}_{t,j}' : g_{t,j}\left(\boldsymbol{x}, \boldsymbol{\theta}_j, \boldsymbol{X}_{klt}^{(m)} = \left\{\boldsymbol{x}_{kli}^{(m)}\right\}_{i=S_l+1}^{S_l+N_t}\right)\right\}_{t=1,j=1}^{T=2,d-1}$$

with $N_1 = N^{(m)}$ and $N_2 = N^{(m)} + \Delta N$, and return to Step 6); if Flag $= 2$, the optimal frame model is $\mathcal{M}_{kl}^{\prime *} : g_{t^*,j_{t^*}}\left(\boldsymbol{x}, \boldsymbol{\theta}_{j_{t^*}}, \boldsymbol{X}_{klt^*}^{(m)}\right)$, and the algorithm ends.

Step 5) Length decrease judgement: If $t^* = 1$, let Flag $= 0$; if $t^* = 2$ and $N^{(m)} - 2\Delta N \ge d$, let Flag $= -1$; if $t^* = 2$ and $N^{(m)} - 2\Delta N < d$, let Flag $= 0$. Return to Step 4) after the judgement.

Step 6) Length increase judgement: If $t^* = 1$, let Flag $= 0$; if $t^* = 2$ and $N^{(m)} + 2\Delta N \le N_{\text{all}}$, let Flag $= 1$; if $t^* = 2$ and $N^{(m)} + 2\Delta N > N_{\text{all}}$, let Flag $= 2$. Return to Step 4) after the judgement.

Based on the AIC and the BIC, the above algorithm can automatically give the optimal aspect-frame boundaries and determine the optimal number of factors in each aspect-frame.

C. Experimental Results Based on Measured Data

According to the introduction in Section II-C-1), by the equal interval partition approach, there are 35 aspect-frames for Yak-42, 50 for Cessna Citation S/ II, and 50 for An-26, of which each frame has 1024 HRRP samples. The proportions of the number of samples in an aspect-frame to the number of all training samples are nearly 3%, 2%, and 2% for Yak-42, Cessna Citation S/ II and An-26, respectively. Fig. 9(a)–(c) shows all of the proportions for Yak-42, Cessna Citation S/ II and An-26 by the adaptive partition approach based on the AIC with the frame step $\Delta N = 4$, in which 32 aspect-frames for Yak-42, 39 for Cessna Citation S/ II, and 48 for An-26; while Fig. 9(d)–(f) shows those by the adaptive partition approach based on the BIC with the frame step $\Delta N = 4$, in which 31 aspect-frames for Yak-42, 35 for Cessna Citation S/ II, and 47 for An-26. Obviously, the lengths of aspect-frames are different, and some are not within the limitation of avoiding scatterers' MTRC, which cannot be allowed in the ISAR imaging. In addition, the aspect-frames in the training data by the adaptive partition approach are fewer than those by the equal interval partition approach, therefore, the computation burden can be reduced. Table IV shows the confusion matrices and average recognition rates obtained by the AIC- and BIC-based equal interval partition approach and the AIC- and BIC-based adaptive partition approach with $\Delta N = 4$. Based on the AIC, the average recognition rate obtained by the adaptive partition approach is approximately 1 percentage point larger than that obtained by the equal interval partition approach; while based on the BIC, the average recognition rate obtained by the adaptive partition approach is approximately 2 percentage points larger than that obtained by the equal interval partition approach. Therefore, the adaptive partition approach further improves the recognition rate with higher recognition efficiency. The numbers of the factors in all aspect-frames by the four aspect-frame partition approaches are shown in Fig. 10. Since the number of samples in an aspect-frame $N = 1024 > 8$ in the equal interval partition approach and the minimum number of samples in an aspect-frame $N_{\min} \ge d = 256 > 8$ in the adaptive partition approach, the BIC favors models with fewer parameters than the AIC does. Fig. 10 demonstrates this point. As shown in Table IV, the average recognition rates obtained by the BIC-based aspect-frame partition approaches are 2 to 3 percentage points larger than those obtained by the AIC-based

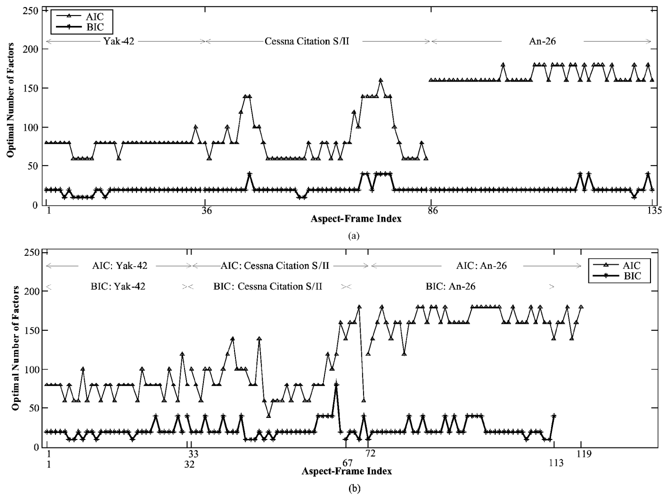

Fig. 10. Numbers of the factors in all aspect-frames by the AIC and BIC-based equal interval partition approach and by the AIC- and BIC-based adaptive partition approach with $\Delta N = 4$. (a) Equal interval partition approach. (b) Adaptive partition approach.

approaches; therefore, the BIC is more suitable for radar HRRP statistical recognition.

IV. Conclusion

From the theoretical analysis and some experiments for the subspace approximation model, PPCA model, and FA model, which are first introduced into radar HRRP statistical recognition, we can have the conclusion that a jointly non-Gaussian distributed HRRP frame approximately follows the joint-Gaussian distribution described by FA model. Therefore, we can apply FA model rather than a joint-Gaussian mixture model to radar HRRP statistical recognition, and the difficulty in statistical modeling for HRRP samples is largely reduced. In addition, this paper proposes an adaptive algorithm based on the AIC and the BIC for model selection of FA model in radar HRRP statistical recognition, which can automatically give the optimal aspect-frame boundaries and determine the optimal number of factors in each aspect-frame. The recognition experiments based on measured data show that the proposed adaptive partition approach can further improve the recognition performance with higher recognition efficiency.

It is worth pointing out that for radar HRRP statistical recognition, a large amount of training data are required to learn the statistical characteristics of target HRRP, therefore, the performance evaluation results will be sounder if more targets and more training data are concerned.

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References

[1] R. A. Mitchell and R. Dewall, "Overview of high range resolution radar target identification," presented at the Automatic Target Recognition Working Group, Monterey, CA, 1994.

[2] H.-J. Li and S.-H. Yang, "Using range profiles as feature vectors to identify aerospace objects," *IEEE Trans. Antennas Propag.*, vol. 41, no. 3, pp. 261–268, 1993.

[3] S. Hudson and D. Psaltis, "Correlation filters for aircraft identification from radar range profile," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 3, pp. 1741–1748, 1993.

[4] A. Zyweck and R. E. Bogner, "Radar target classification of commercial aircraft," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 32, no. 2, pp. 598–606, 1996.

[5] R. A. Mitchell, "Robust high range resolution radar target identification using a statistic feature based classifier with feature level fusion," Ph.D. dissertation, Elect. Eng. Dept., University of Dayton, Dayton, OH, 1997.

[6] S. P. Jacobs, "Automatic target recognition using high-resolution radar range profiles," Ph.D. dissertation, Elect. Eng. Dept., Washington Univ., St. Louis, MO, 1999.

[7] R. Van der Heiden and F. C. A. Groen, "The box-cox metric for nearest neighbour classification improvement," *Pattern Recogn.*, vol. 30, no. 2, pp. 273–279, 1997.

[8] R. Williams, J. Westerkamp, D. Gross, and A. Palomino, "Automatic target recognition of time critical moving targets 1-D high range resolution (HRR) radar," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 15, no. 4, pp. 37–43, 2000.

[9] L. Du, H.-W. Liu, and Z. Bao, "Radar HRRP target recognition based on higher-order spectra," *IEEE Trans. Signal Process.*, vol. 53, no. 7, pp. 2359–2368, Jul. 2005.

[10] K. T. Kim, D. K. Seo, and H. T. Kim, "Efficient radar target recognition using the MUSIC algorithm and invariant feature," *IEEE Trans. Antennas Propag.*, vol. 50, no. 3, pp. 325–337, 2002.

[11] X.-J. Liao, P. Runkle, and L. Carin, "Identification of ground targets from sequential high-range-resolution radar signatures," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, no. 4, pp. 1230–1242, 2002.

[12] Y. Shi and X.-D. Zhang, "A G atom network for signal classification with application in radar target recognition," *IEEE Trans. Signal Process.*, vol. 49, no. 12, pp. 2994–3004, Dec. 2001.

[13] A. R. Webb, "Gamma mixture models for target recognition," *Pattern Recogn.*, vol. 33, pp. 2045–2054, 2000.

[14] K. Copsey and A. R. Webb, "Bayesian gamma mixture model approach to radar target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 39, no. 4, pp. 1201–1217, 2003.

[15] L. Du, H.-W. Liu, Z. Bao, and J.-Y. Zhang, "A two-distribution compounded statistical model for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 54, no. 6, pp. 2226–2238, Jun. 2006.

[16] L. Du, H.-W. Liu, Z. Bao, and J.-Y. Zhang, "Radar automatic target recognition using complex high-resolution range profiles," *IET Radar, Sonar, Navigation (formerly Proc. Inst. Elect. Eng.—Radar, Sonar, Navig.)*, vol. 1, no. 1, pp. 18–26, 2007.

[17] L. Du, H.-W. Liu, and Z. Bao, "Using the amplitude fluctuation property of target HRRP for radar automatic target recognition," in *Proc. 2006 Int. Confe. Radar.*, Oct. 2006, pp. 829–833.

[18] W. G. Carrara, R. S. Goodman, and R. M. Majewski, *Spotlight Synthetic Aperture Radar -Signal Processing Algorithms*. Norwood, MA: Arthech House, 1995.

[19] M. E. Tipping and C. M. Bishop, "Mixtures of principal component analysers," *Neural Comput.*, vol. 11, no. 2, pp. 443–482, 1999.

[20] D. Rubin and D. Thayer, "EM algorithms for ML factor analysis," *Psychometrika*, vol. 47, no. 1, pp. 69–76, 1982.

[21] Z. Ghahramani and G. E. Hinton, "The EM algorithm for mixtures of factor analyzers," Dept. Comp. Sci., Univ. of Toronto, Toronto, Canada, Tech. Rep. CRG-TR-96-1, 1996.

[22] S. Sclove, "Application of model-selection criteria to some problems in multivariate analysis," *Psychometrika*, vol. 52, no. 3, pp. 333–343, 1987.

[23] H. Akaike, "Factor analysis and AIC," *Psychometrika*, vol. 52, no. 3, pp. 317–332, 1987.

[24] P. M. Djurić, "Asymptotic MAP criteria for model selection," *IEEE Trans. Signal Process.*, vol. 46, no. 10, pp. 2726–2735, Oct. 1998.

[25] R. O. Duda, P. E. Hart, and D. G. Stork, *Pattern Classification*, 2nd ed. New York: Wiley, 2001.

[26] H. Attias, "Independent factor analysis," *Neural Comput.*, vol. 11, no. 4, pp. 803–851, 1999.

[27] S. Ikeda and K. Toyama, "Independent component analysis for noisy data—MEG data analysis," *Neural Netw.*, vol. 13, pp. 1063–1074, 2000.

Lan Du received the M.Eng. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in 2004 and 2007, respectively.

She is currently a visiting scholar with the Department of Electrical and Computer Engineering, Duke University, Durham, NC . Her main research interests are in the fields of statistical signal processing, radar signal processing, machine learning, and its application to radar automatic target recognition (RATR).

Hongwei Liu (M'04) received the M.S. and Ph.D. degrees, both in electronic engineering, from Xidian University, Xi'an, China, in 1995 and 1999, respectively.

He worked at the National Laboratory of Radar Signal Processing, Xidian University, after that. From 2001 to 2002, he was a visiting scholar at the Department of Electrical and Computer Engineering, Duke University, Durham, NC. He is currently a Professor at the National Laboratory of Radar Signal Processing, Xidian University. His research interests are radar automatic target recognition, radar signal processing, and adaptive signal processing.

Zheng Bao (M'80–SM'90) received the B.Eng. degree from the Communication Engineering Institution of China in 1953.

Currently, he is a Professor at Xidian University, Xi'an, China. He is the author or coauthor of six books and has published more than 300 papers. His current research work focuses on the areas of space–time adaptive processing (STAP), radar imaging (SAR/ISAR), radar automatic target recognition (RATR), over-the-horizon radar (OTHR) signal processing, and passive coherent location (PCL).

Prof. Bao is a member of the Chinese Academy of Science.